

Industrial AI & Research Roadmap for JSPL EAF

Prepared as Phase 27 deliverable for management and academic planning.

A. Next 12 months (plant)

- 1 Instrument P0: delay codes, power-on/off, charging/crane waits, restriction flag.
- 2 Shadow-deploy Phase 25 two-stage + Phase 26 gold features; do not replace Phase 19 yet.
- 3 Confirm OXY/CPC are planning setpoints vs final totals (Phase 23.5 open question).
- 4 Define KPI dashboard: normal MAE, delay recall, false-alarm rate.

B. Next 3 years (digital twin)

- 1 Year 1–2: P1 quality (metallization, HM temp) and gas injection profiles; Stage-1 recall >80%.
- 2 Year 2–3: V3 twin with live residual monitoring; electrode/alarm analytics.
- 3 Year 3: V4 closed-loop optimizer with human confirmation; EE_KWH used only as outcome KPI.

C. Recommended MSc / PhD topics

Topic	Degree fit	Data need
Dynamic TTT prediction with streaming SCADA	MSc / PhD	Power-on, V/I
Causal delay prediction from MES event logs	MSc	Delay codes
Physics-informed slag foam & energy models	PhD	Slag + C/O2 profiles
Two-stage / hierarchical regime models for EAF	MSc	Current + events
Safe RL for recipe adaptation under constraints	PhD	Twin sandbox
Predictive maintenance from electrode & alarms	MSc	Historian
Digital twin validation vs plant KPI	PhD	Full stack
DRI quality → melt time transfer learning	MSc	Metallization

Evidence basis: Phase 19 MAE≈3.06; Phase 25 LF normal≈3.28; Phase 26 best≈3.24; mixed-cohort single model≈36.0; two-stage plant MAE≈16.0; delay recall≈22%.